

ULIASTAI, Mongolia

WMO: 442720

Lat: 47.7272N

Lon: 96.8475E

Elev: 5771

StdP: 11.88

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-30.5	-26.8	-36.3	0.9	-29.8	-33.0	1.1	-25.9	14.4	5.6	11.5	-2.2	2.3	180	0.660

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.0	82.3	58.2	79.0	56.9	75.9	55.4	61.1	76.7	59.0	74.4	57.2	71.7	7.5	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
55.7	82.0	64.1	53.2	74.8	64.0	51.0	68.8	63.4	30.1	76.4	28.5	74.4	27.2	71.9	70.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.7	14.1	11.9	-35.1	87.7	4.9	3.5	-38.7	90.3	-41.5	92.3	-44.3	94.2	-47.9	96.8
			-35.1	63.7	4.9	3.2	-38.7	66.0	-41.5	67.9	-44.3	69.6	-47.9	71.9

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	29.9	-9.0	-1.3	15.9	36.7	47.6	58.8	63.1	58.9	47.0	31.7	11.0	-3.6	(6)
(7)		DBStd	26.68	9.57	10.63	12.27	8.97	7.96	6.39	5.67	6.03	7.14	9.02	11.64	8.96	(7)
(8)		HDD50	8413	1830	1437	1057	401	138	10	0	6	137	567	1169	1661	(8)
(9)		HDD65	12889	2295	1857	1522	848	539	201	105	204	541	1032	1619	2126	(9)
(10)		CDD50	1074	0	0	0	3	64	273	406	281	47	0	0	0	(10)
(11)		CDD65	74	0	0	0	0	0	14	45	15	0	0	0	0	(11)
(12)		CDH74	1079	0	0	0	0	22	247	579	226	5	0	0	0	(12)
(13)		CDH80	210	0	0	0	0	0	35	141	34	0	0	0	0	(13)
(14)	Wind	WSAvg	3.6	2.1	2.3	3.3	4.7	5.3	4.8	4.3	4.0	4.1	3.3	2.4	2.2	(14)
(15)	Precipitation	PrecAvg	7.8	0.1	0.1	0.2	0.4	0.6	1.3	2.3	1.7	0.8	0.4	0.2	0.1	(15)
(16)		PrecMax	12.9	0.4	0.4	0.8	1.3	1.9	3.5	7.0	3.7	3.0	1.2	0.6	0.4	(16)
(17)		PrecMin	3.5	0.0	0.0	0.0	0.0	0.0	0.2	0.4	0.0	0.0	0.0	0.0	0.0	(17)
(18)		PrecStd	2.1	0.1	0.1	0.2	0.3	0.5	0.7	1.4	0.9	0.6	0.3	0.1	0.1	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	21.1	30.0	50.5	66.1	76.8	84.4	88.7	83.9	73.9	60.0	44.1	25.8	(19)
(20)		0.4%	MCWB	17.4	24.7	37.0	47.3	53.0	58.2	62.2	58.6	52.0	43.8	33.8	21.4	(20)
(21)		2%	DB	14.1	23.5	44.1	61.1	72.5	79.8	83.9	79.7	69.2	54.9	38.3	17.5	(21)
(22)		2%	MCWB	11.9	19.5	33.1	44.0	51.4	56.8	59.7	56.8	50.7	40.8	30.1	15.0	(22)
(23)		5%	DB	9.4	19.1	39.4	56.7	68.4	76.8	80.3	76.1	65.5	51.3	33.2	13.7	(23)
(24)		5%	MCWB	7.4	15.8	30.6	41.6	49.1	55.5	57.5	55.9	48.5	38.5	26.6	11.5	(24)
(25)		10%	DB	5.2	15.1	34.8	52.7	64.2	73.7	76.8	72.7	61.7	46.9	28.0	9.8	(25)
(26)		10%	MCWB	3.5	12.3	27.9	38.9	46.3	53.9	56.3	54.5	46.9	36.1	23.2	7.9	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	17.7	25.1	37.6	48.0	56.1	62.0	65.4	61.3	54.7	45.0	34.9	21.2	(27)
(28)		0.4%	MCDWB	20.7	29.8	49.6	62.4	75.1	79.1	79.5	77.2	69.0	57.9	43.2	25.1	(28)
(29)		2%	WB	12.0	19.6	33.7	45.0	52.2	58.9	62.4	59.2	51.8	41.8	30.5	15.3	(29)
(30)		2%	MCDWB	14.0	23.4	42.8	59.3	69.3	76.4	79.4	73.6	65.5	53.3	37.8	17.5	(30)
(31)		5%	WB	7.5	15.9	31.0	42.3	49.6	56.9	60.2	57.4	49.8	39.4	26.9	11.5	(31)
(32)		5%	MCDWB	9.4	19.0	39.0	55.4	66.3	73.1	75.3	71.3	62.8	50.4	33.2	13.5	(32)
(33)		10%	WB	3.6	12.4	28.1	39.6	47.2	55.1	58.2	55.7	47.7	36.9	23.6	7.9	(33)
(34)		10%	MCDWB	5.1	15.2	34.8	51.5	62.7	69.7	72.8	69.6	60.4	47.0	28.3	9.8	(34)
(35)	Mean Daily Temperature Range		MDBR	18.3	22.9	25.4	21.4	23.2	22.7	21.0	21.9	24.2	23.7	19.9	16.9	(35)
(36)		5% DB	MCDBR	20.7	24.6	25.2	27.2	29.8	27.7	27.3	27.7	29.3	29.2	25.4	19.2	(36)
(37)			MCWBR	18.7	21.5	17.4	15.5	16.4	13.3	11.9	12.4	15.7	18.0	18.1	17.2	(37)
(38)			MCDBR	20.5	24.6	24.4	25.6	27.8	24.9	23.1	23.6	25.9	27.8	24.4	19.2	(38)
(39)		5% WB	MCWBR	18.6	21.4	17.2	15.3	16.5	13.0	11.8	11.5	14.6	17.7	17.9	17.3	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.225	0.253	0.296	0.339	0.338	0.349	0.348	0.326	0.292	0.274	0.248	0.229	(40)	
(41)		taud	2.336	2.268	2.209	2.232	2.294	2.353	2.390	2.424	2.473	2.438	2.380	2.325	(41)	
(42)		Ebn at Noon	281	292	293	289	293	289	288	291	293	282	267	263	(42)	
(43)		Edh at Noon	22	30	37	41	40	38	36	33	29	25	21	19	(43)	
(44)	All-Sky Solar Radiation	RadAvg	561	904	1399	1819	2137	2149	2035	1805	1517	988	568	440	(44)	
(45)		RadStd	43	56	52	75	130	106	123	80	68	65	44	32	(45)	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (5 neighbors)	N/A	N/A	-3.51	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page